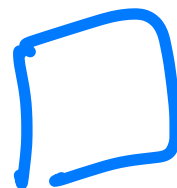
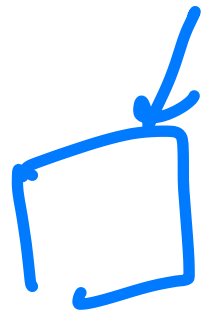


Goal $\rightarrow$ Autoregressive?

Compressed $\rightarrow$ Decompressed /
Reconstructed
image


original
image

$\approx$



~~CNN~~

$\Rightarrow$ ANN

$$z = \underbrace{w_1 x_1 + w_2 x_2 + w_3 x_3 \dots}_{\text{Neuron}} + \underbrace{b}_{\text{Neuron}}$$

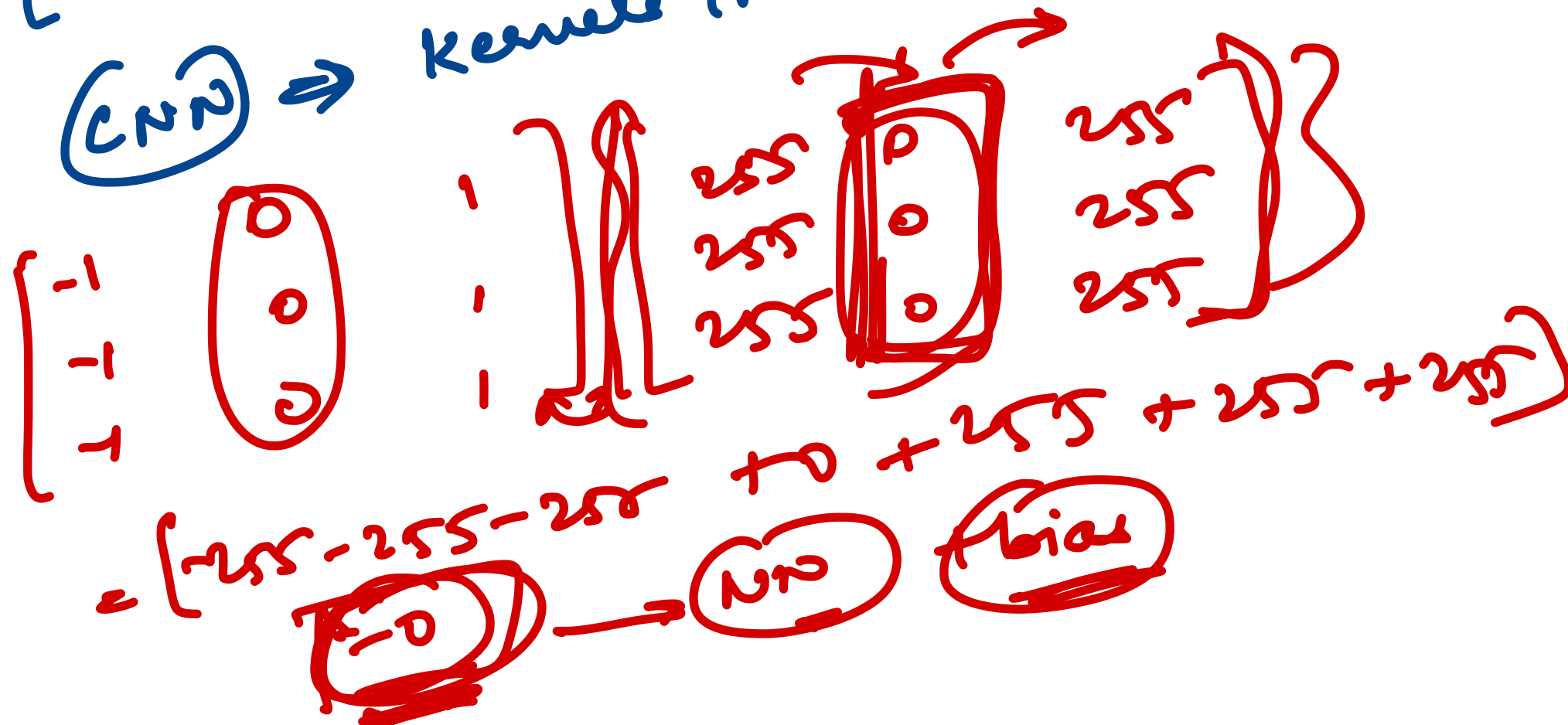
Activation function $f(z)$.

Filters / Kernel

Neuron



(CNN) $\Rightarrow$ Kernel / Filters $\rightarrow$ features



ANN $\rightarrow$ learn
params

CNN $\rightarrow$ "

$\rightarrow w \& b$
 $\rightarrow$ " $\downarrow$
elements
in kernel

detecting
all features

image:

$$\begin{bmatrix} x_1 & & & \\ & x_2 & & \\ & & x_3 & \\ x_4 & & & \\ & x_5 & & \\ & & x_6 & \\ x_7 & & & x_8 \end{bmatrix} \begin{bmatrix} w_0 & w_1 & w_2 \\ w_3 & w_4 & w_5 \\ w_6 & w_7 & w_8 \end{bmatrix}$$

$$\{ w_0 x_1 + w_0 x_2 + w_2 x_3 - \dots - + b \}$$

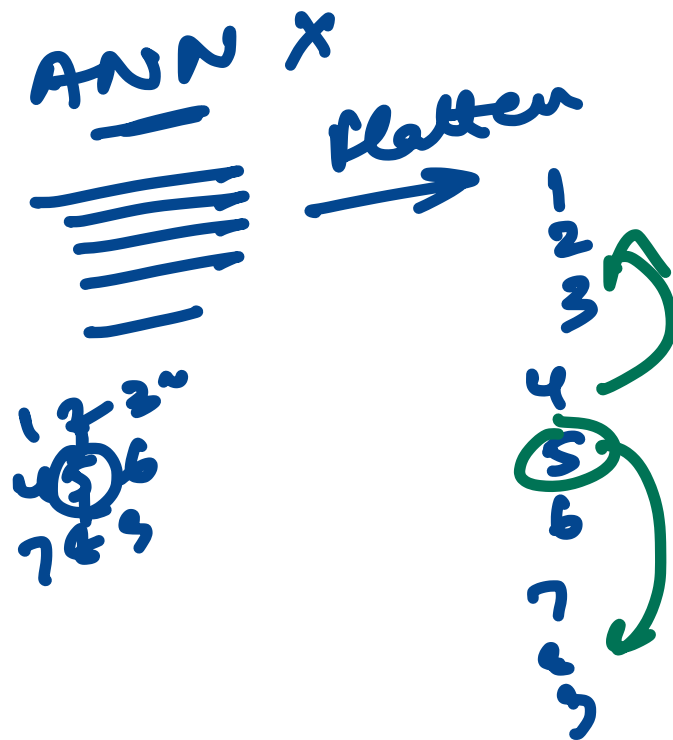
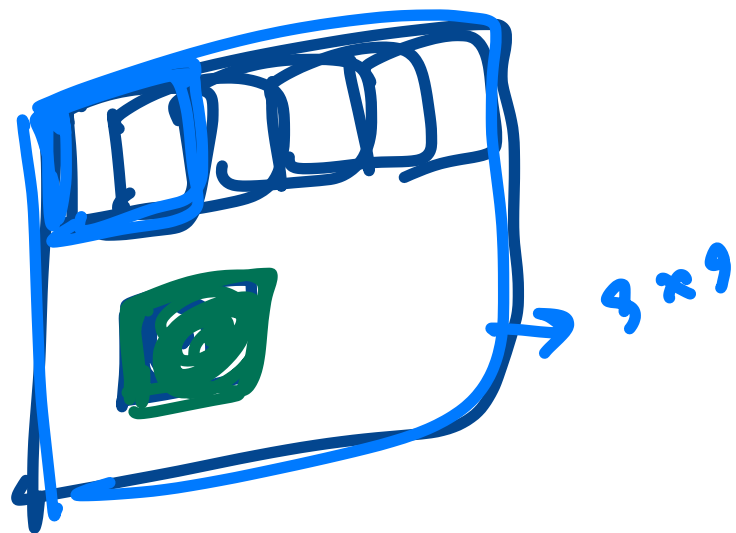
$\downarrow$
ANN

$\downarrow$
dot product
weighted sum of
our input

Neuron $\rightarrow$ Filter / Kernel
weights $\rightarrow$ connects $\rightarrow$ elements of filters

Diff
ANN $\rightarrow$ Single o/p $\xrightarrow{\text{o/p}}$ Activation
 $\Rightarrow$ Feature Map $\rightarrow$ Matrix

CNN $\rightarrow$ Patterns



Encoder $\rightarrow$

$$\underline{28 - 28 = 1}$$



$$\rightarrow \text{O/P } (14, 14, 32) \downarrow \text{feature maps}$$

① conv 2D layer

32 filters

kernel $\Rightarrow 3 \times 3$

Stride = 2

Per filter, no. of parameters

$$\underbrace{3 \times 3}_{\text{wts}} + \underbrace{1}_{\text{bias}}$$

= 10 params

32 filters $\Rightarrow$ 10 params $\Rightarrow$ 320 params

② conv 2D layer 2

64 filters

ilp $\Rightarrow$ 32 feature maps
64 filters for this layer

Per Filter

No. of params

$$(32 * 3 * 3) + 1 = 288 + 1 = 289$$

64 Filters

$$\rightarrow 64 * 289 =$$

$$18,496$$

$$o/p \Rightarrow 7, 7, 64$$

$$\Downarrow$$
$$7 * 7 * 64 = 3136$$

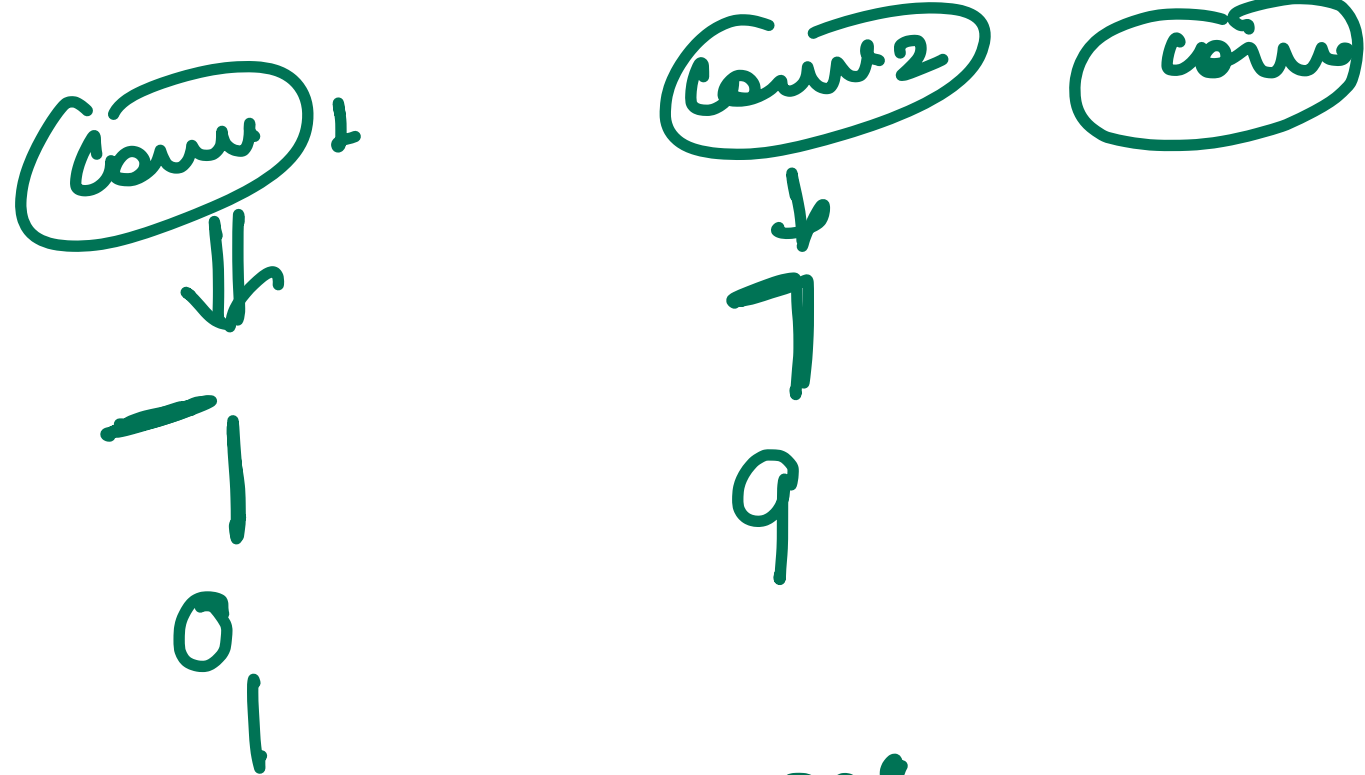
③ Flatten layer (Params)
0

④ Dense layer ID
o/p $\Rightarrow 64$

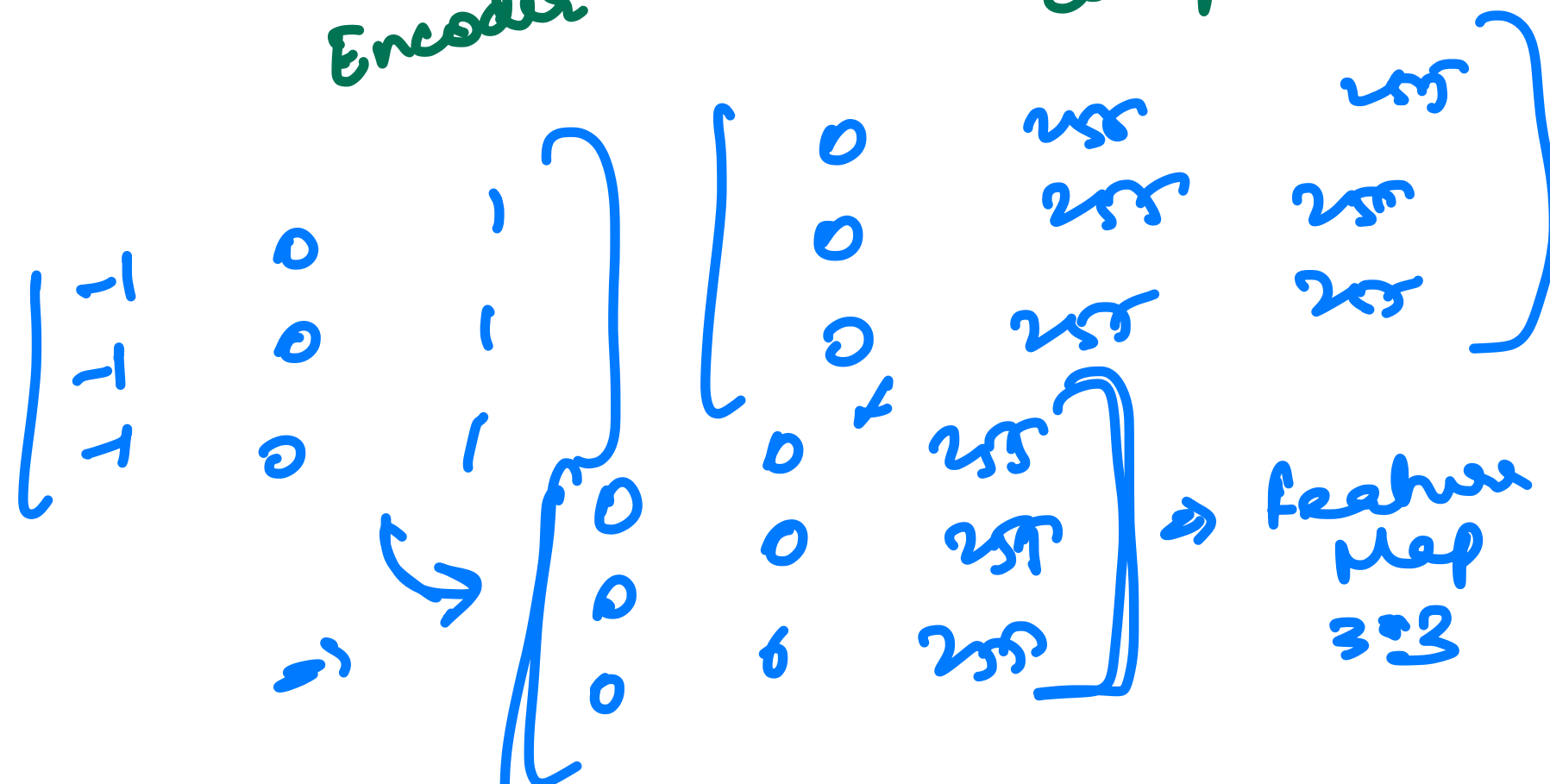
$$3136 + 1 \text{ bias per neuron} \Rightarrow 3137 \text{ params}$$

$$64 \text{ neurons} * 3137 \text{ params} =$$

$$200768$$



Cat dog
Detecting feature → Count
Encoder → o/p → 1D vector compressed



o/p $\Rightarrow 3 \times 3$ FM
of $(14, 14)$
Kernel $\Rightarrow 3 \times 3$

$$3 \times 3 = 32$$

Decoder

dense $\Rightarrow 64 \Rightarrow 3136$
Reshape

$(7, 7, 64)$

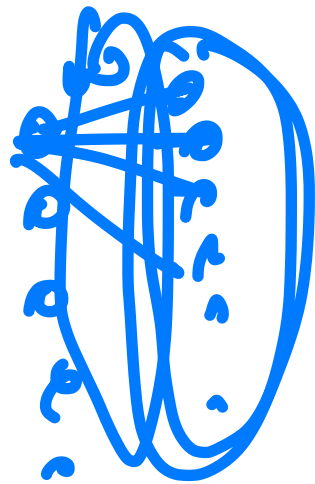
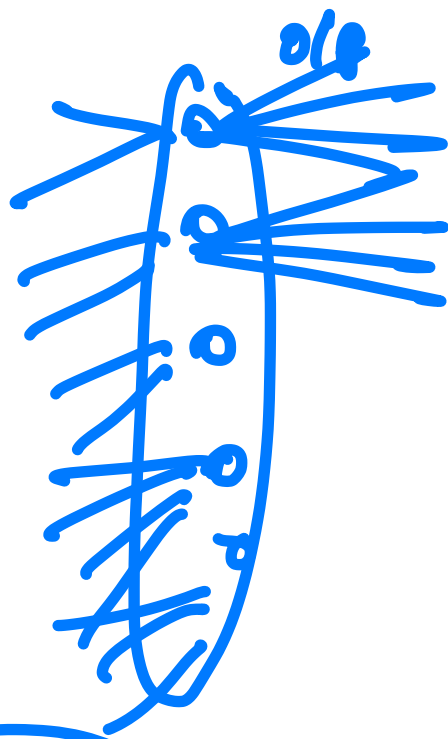
$\Downarrow$
 $14, 14,$

$28, 28$

$\Downarrow$
 $(28, 28, 1)$ o/p

Decoder

① Dense $(64+1) \times 3136$
 65×3136



203840

$(7, 7, 64)$

② Reshape

③ conv 2D Transpose

Filter $\rightarrow 64$

kernel $\rightarrow (3, 3)$

o/p $\rightarrow 14, 14, 64$

$\rightarrow$ upsampling

$(3 \times 3 \times 64 \times 64) + 64$
 $\downarrow$
 wts filter bias

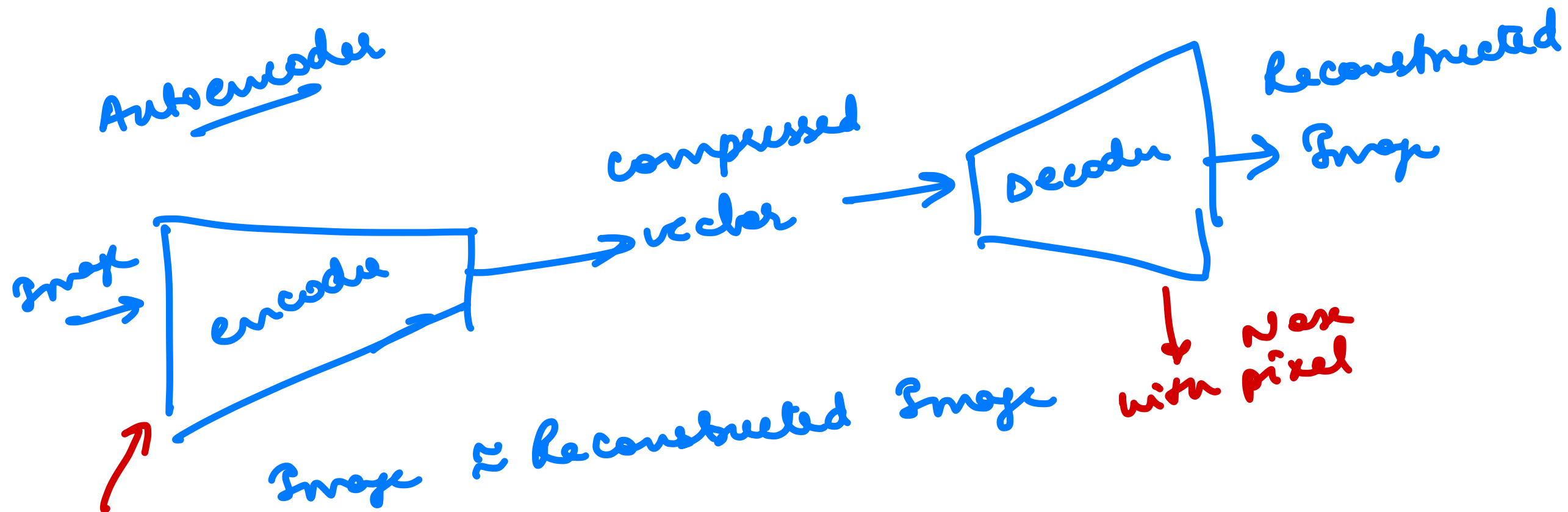
$(\text{Filter width} \times \text{Filter height} \times \text{no. of feature maps} \times \text{2D channels}) + \text{bias}$
 $\approx \text{o/p channels}$

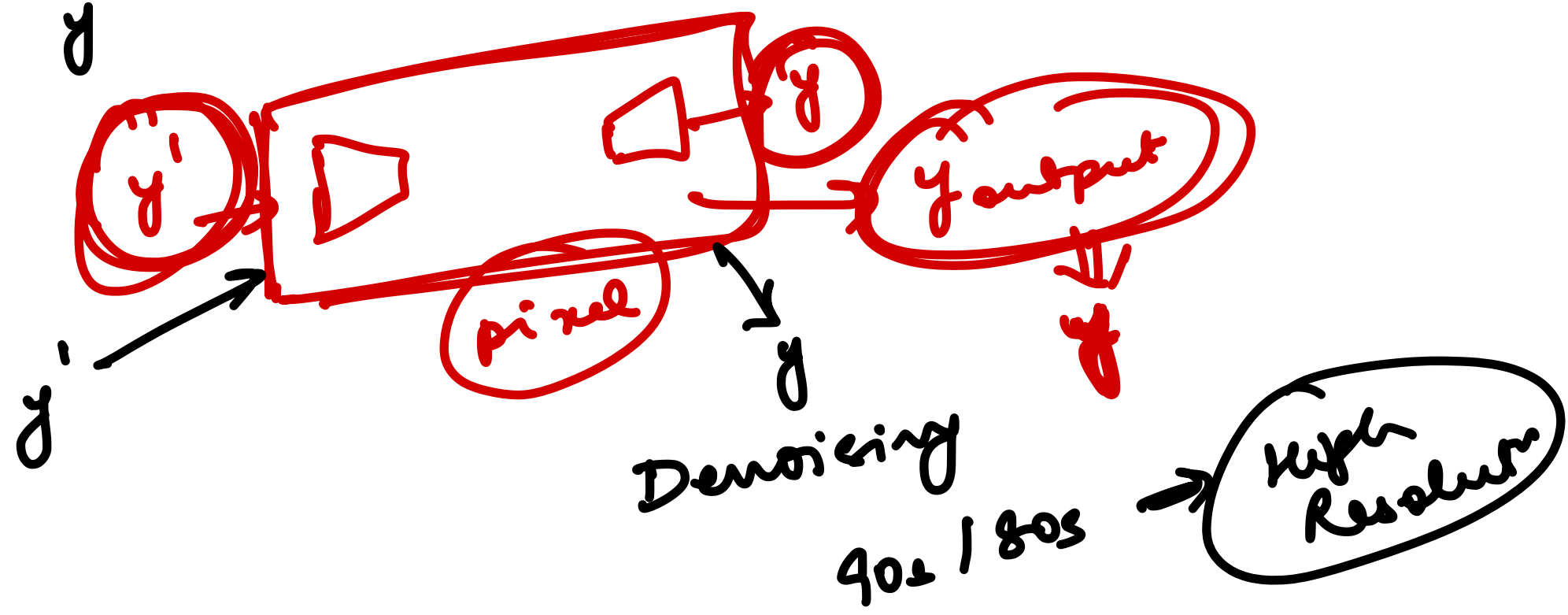
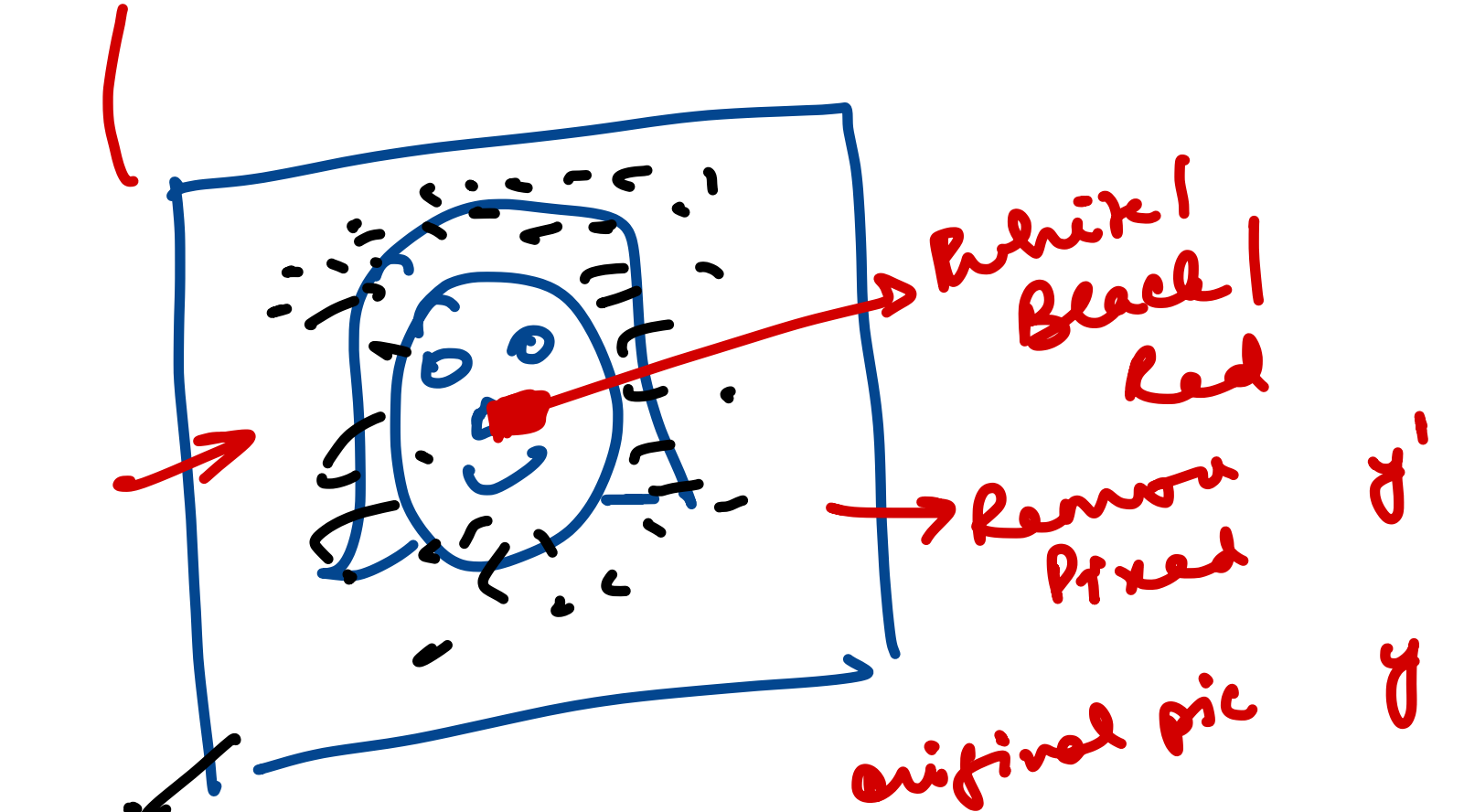
$$(3 \cdot 3) \cdot 64 = 32 + 32$$

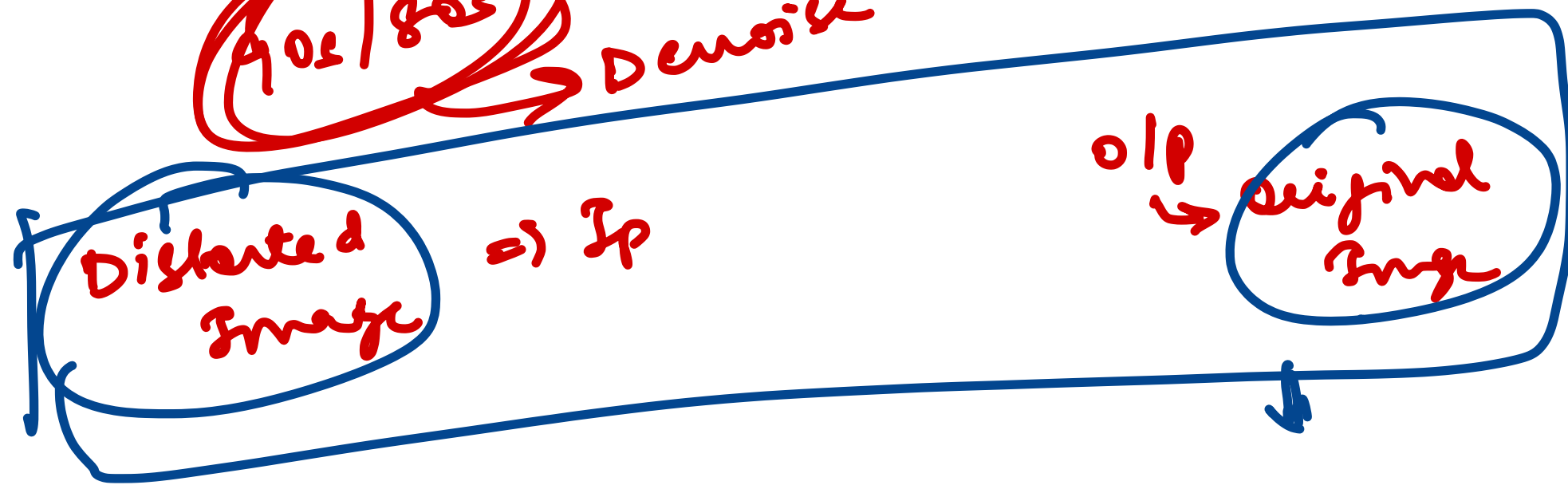
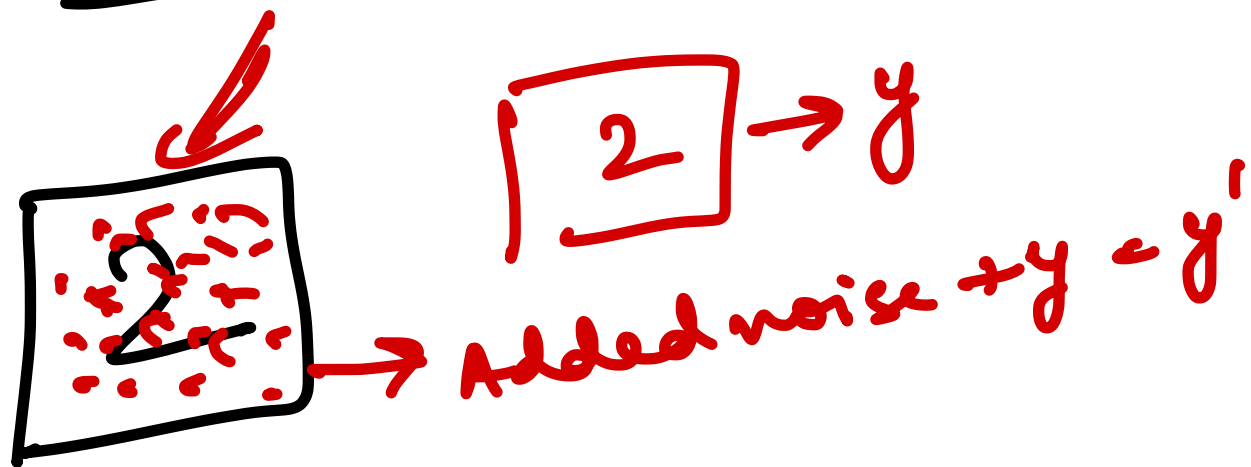
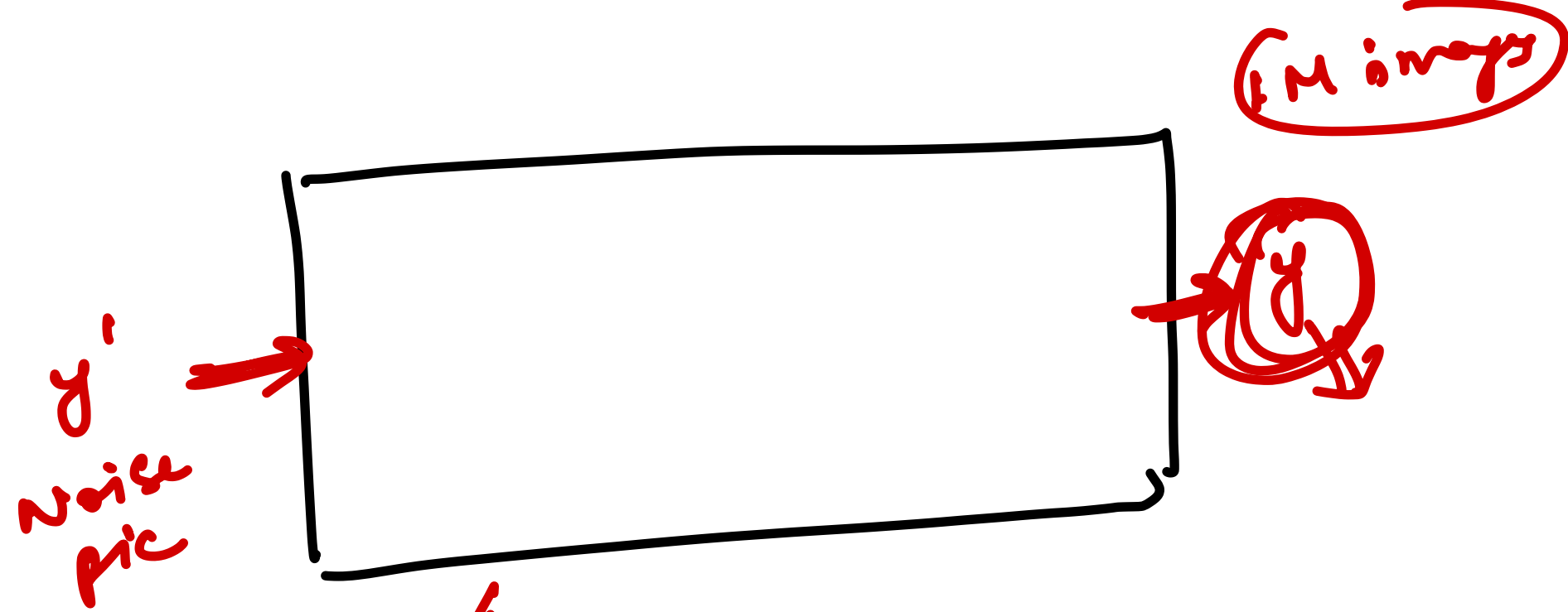
↓ ↓
filter bias

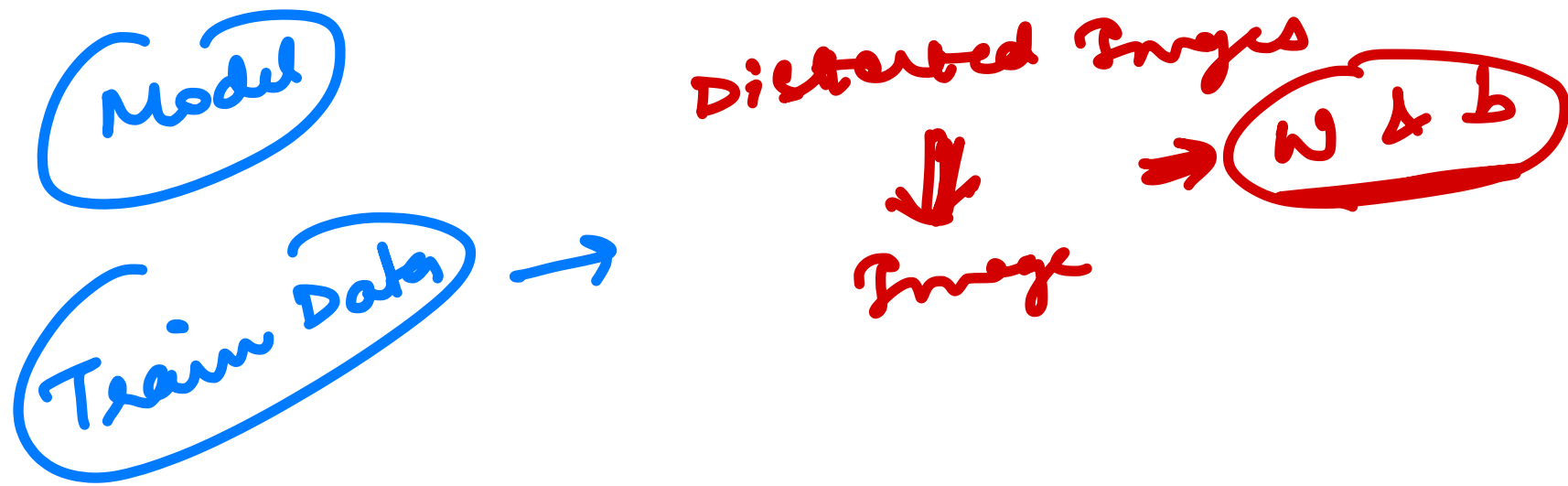
$$= 18464$$

Removing watermark
 Remove obj from image
 Auto fill

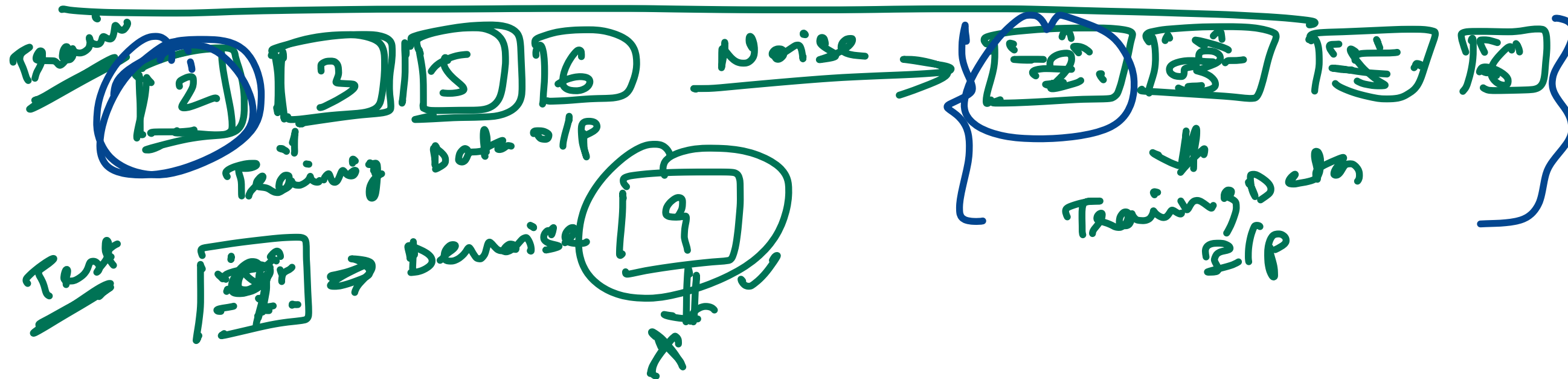
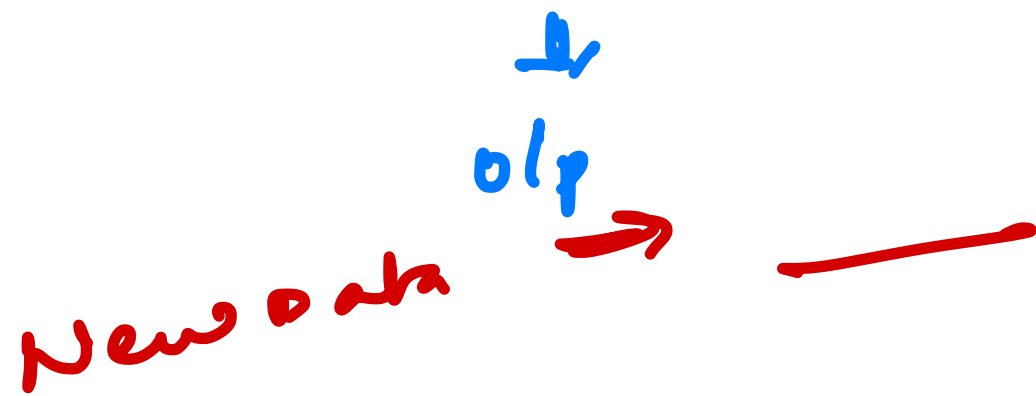


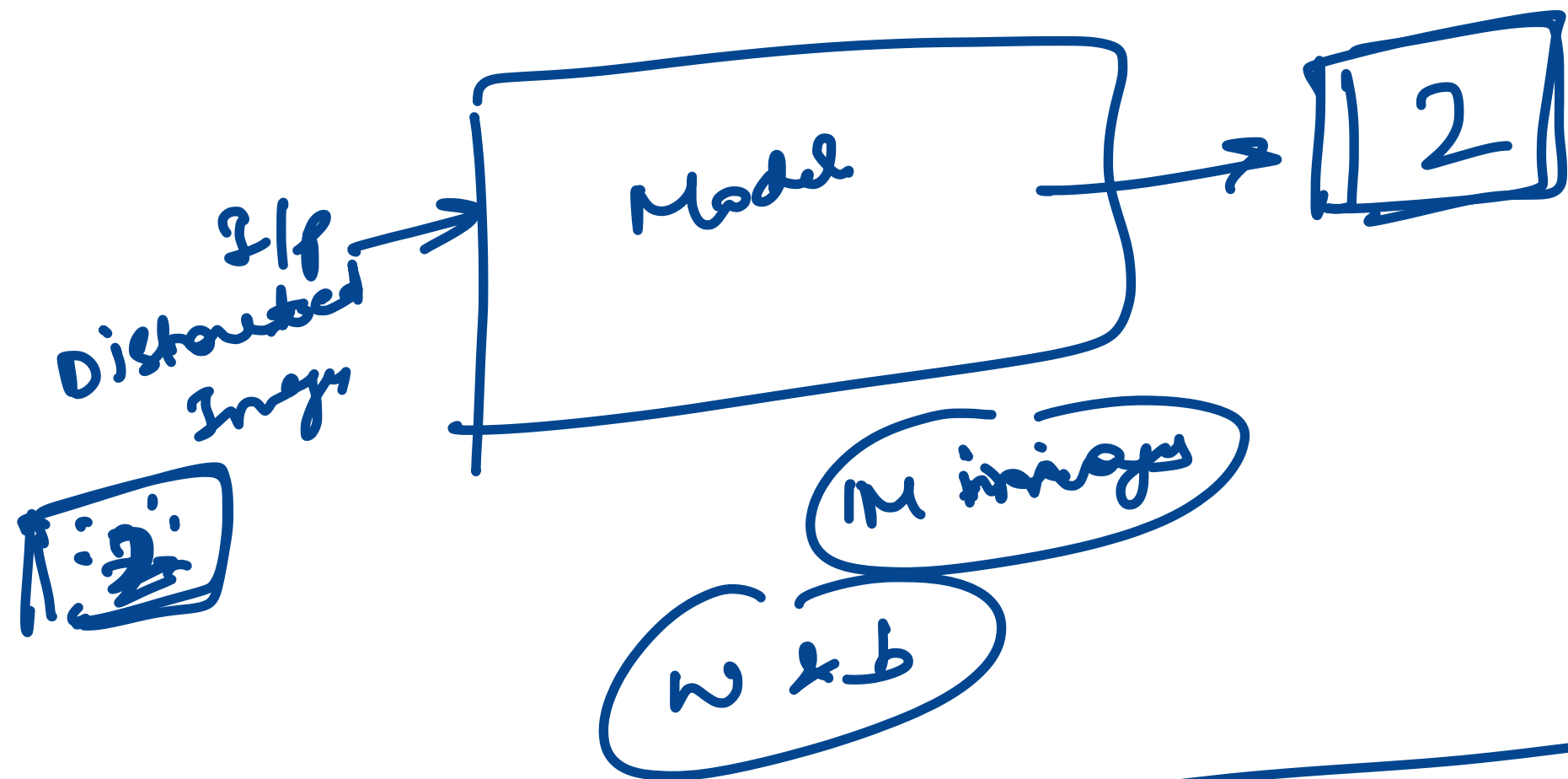






Test Data → New Data





Train over



Generation of New Images

Generational Models

Midjourney, Dall-E
↓
create new images

Deepfake
{ Filters on
Snapchat,
TikTok }

GAN
↓
Generative
Adversarial
Net

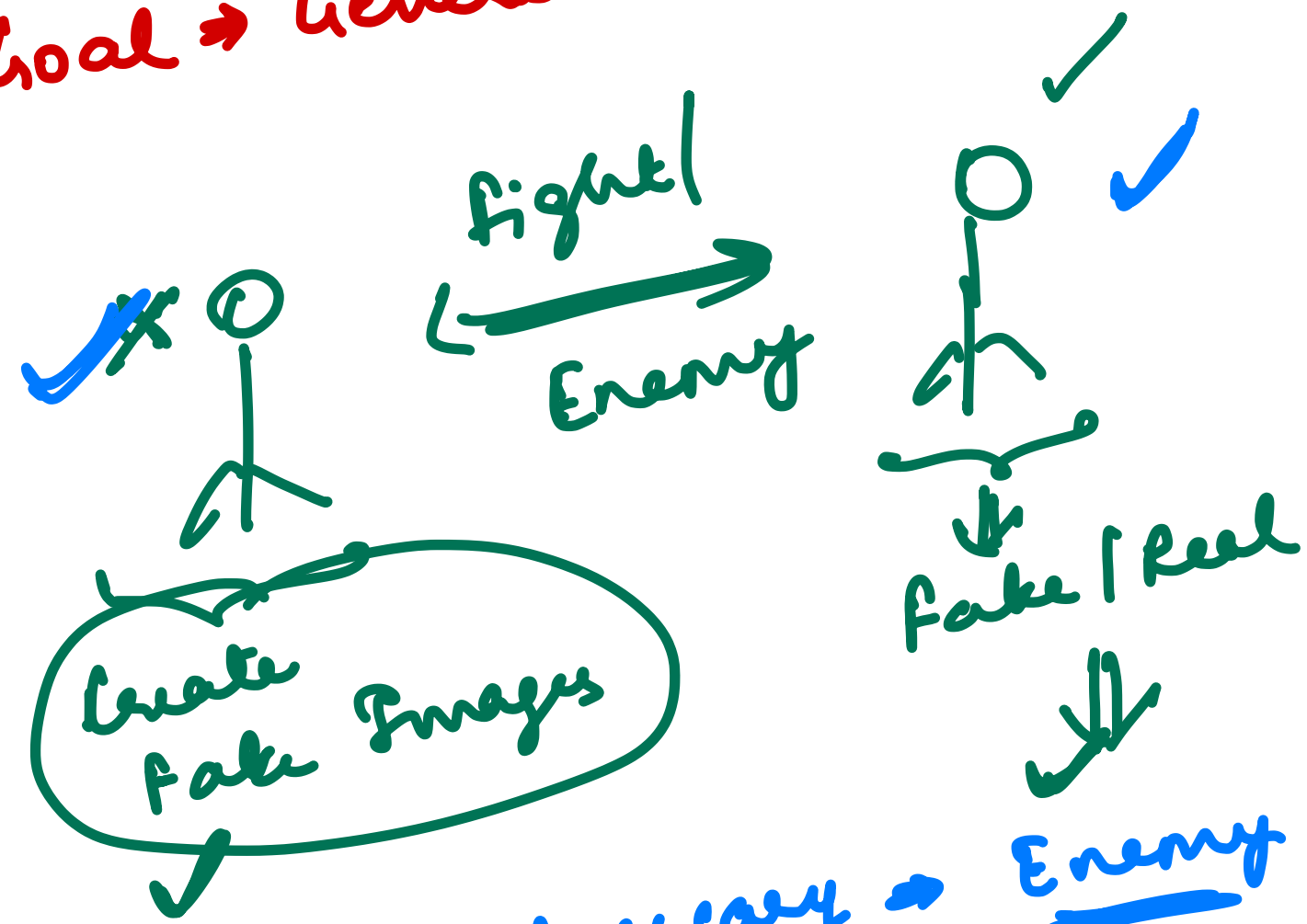
VAE
↓
variational
autoencoders

Diffusion
Model

GANs

videos
audio

Goal $\rightarrow$ Generate New Images



Adversary $\rightarrow$ Energy

2 Models $\rightarrow$ Energy
Fight

① Generator

→ generate
fake images

② Discriminator

→ Discriminate
Between Real
and fake Images



Discriminate

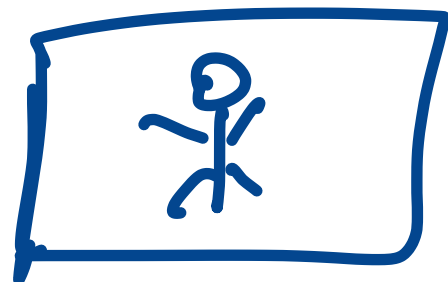
✓ fake ✓

generator → Train
w & b

Real
generator
X

Discriminate
w & b

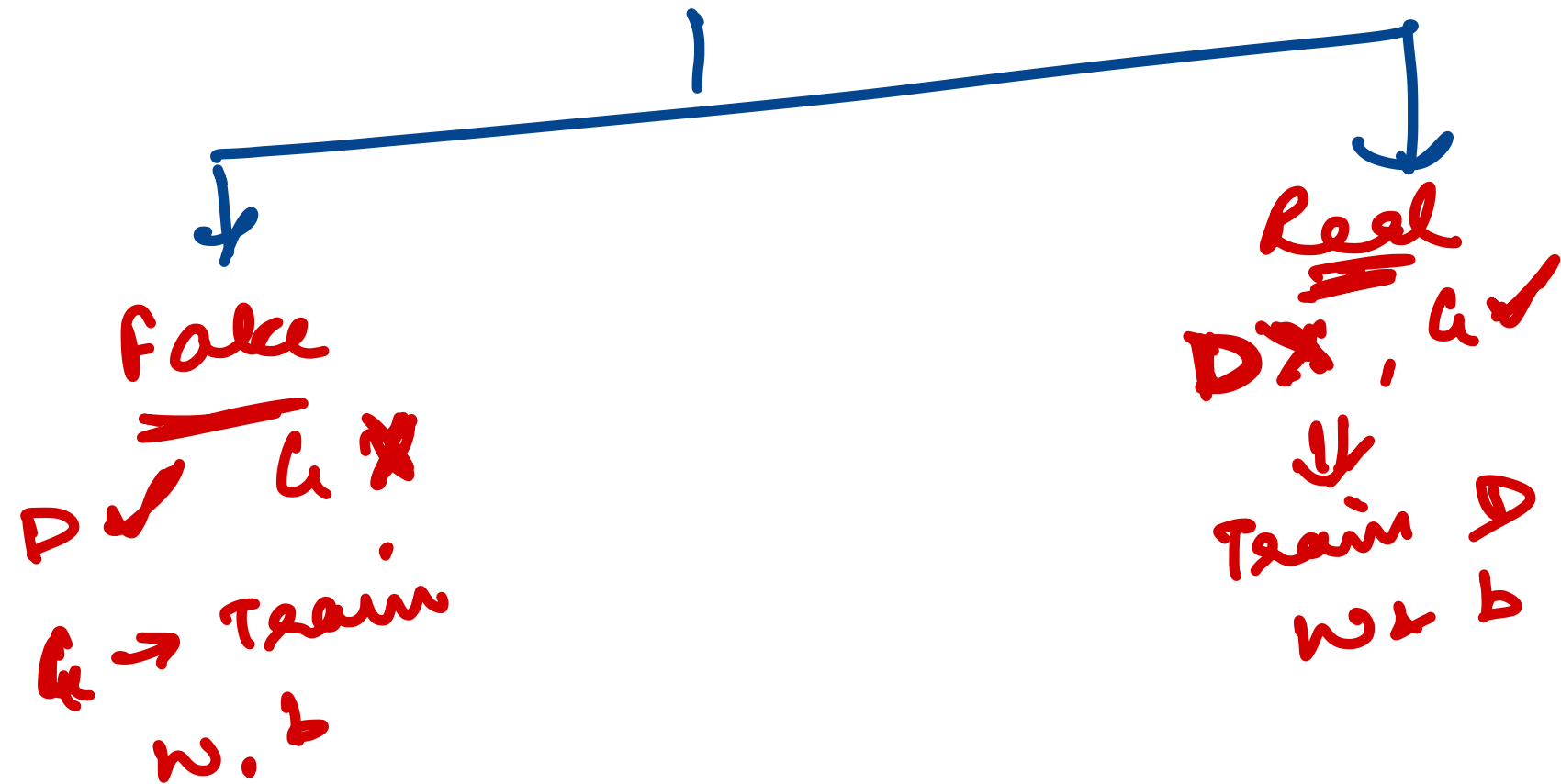
Generator



fake



Discriminator



winner, loser

winner

loser

remain unchanged

update model

→ complete → 3, 3, 3, 3, 3

→ Partial → 3/7/9

→ Intermittent →

[100 imp] → Diversity

Mode Collapse

